Graph Attention in a Learned Branching Heuristic for SAT

GAT-Q-SAT against Graph-Q-SAT, at matched training set and with two controls

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The branching heuristic of a CDCL solver

- A CDCL solver repeatedly decides the value of a variable, propagates, and on a conflict learns a clause and backjumps; the choice of the decision is left to a heuristic, usually VSIDS, which ranks the variables by activity counters.
- The counters carry little information at the start of the search, and a poor decision costs time but never correctness: the heuristic is a natural place for a learned policy.
- Graph-Q-SAT (Kurin et al., 2020) learns it by Deep Q-Learning on the bipartite graph of the formula, and sums the messages of all the neighbors of a node with the same weight.
- This work adds graph attention to its core block (GAT-Q-SAT), and asks where weighting the neighbors helps, and whether the gain comes from the attention or from the added parameters.

The method

- Branching as a decision process
- Graph-Q-SAT
- GAT-Q-SAT
- The two controls

Kurin et al., *Can Q-Learning with Graph Networks Learn a Generalizable Branching Heuristic for a SAT Solver?*
Veličković et al., *Graph Attention Networks.*

Branching as a decision process

- The state is the formula simplified by the partial assignment, with the learned clauses; an action is an unassigned variable with a truth value; a step is one decision with the propagation and the conflict analysis it triggers.
- The reward is -0.1 at each step and 0 at termination, so that maximizing the return means terminating in as few decisions as possible.
- DQN approximates $Q^\star(s,a) = \mathbb{E}\big[r + \gamma \max_{a'} Q^\star(s',a')\big]$ with a target network and a uniform replay buffer, and the heuristic is $\pi(s) = \arg\max_a Q_\theta(s,a)$.
- The number of actions changes along an episode and the state is defined up to a permutation of variables and clauses: hence a graph network, which scores each variable node and applies to formulas of any size.

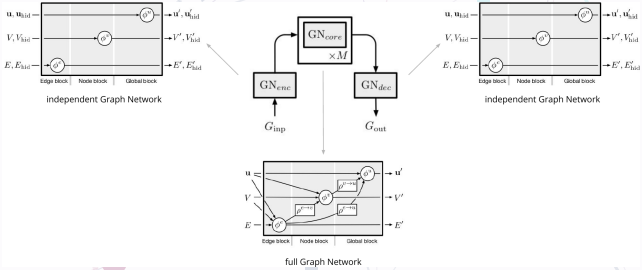
Configuration of Kurin et al.: Adam at $2 \cdot 10^{-5}$, $\gamma = 0.99$, buffer of 20000, minibatches of 64, 50000 updates, at most 500 decisions of the model per episode (the cap).

Graph-Q-SAT

Nodes: the free variables and the clauses not yet satisfied; edges: the occurrences, with the polarity. A core block, applied $M = 4$ times with shared weights, updates the edges, the nodes and the global attribute:

$$e'_{ij} = \phi^e(e_{ij}, v_i, v_j, u)$$
$$v'_i = \phi^v(\sum_{j \in \mathcal{N}_i} e'_{ij}, v_i, u)$$
$$u' = \phi^u(\bar{e}', \bar{v}', u)$$

The sum gives the same weight to all the clauses of a variable; 71144 parameters.



encoder, core block $\times M$, decoder (after Battaglia et al.)

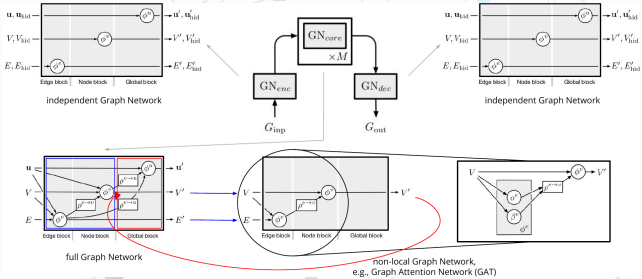
GAT-Q-SAT

Two edge-featured attention layers between the node and the global update (3 heads, then 1):

$$v_i^+ = \Big\|_k \sum_{j \in \mathcal{N}_i} \alpha_{ij}^k W_k v_j + b$$

$$\alpha_{ij}^k \propto \exp \mathrm{LeakyReLU}(\mathbf{w}_k^\top [W_k v_i \,\|\, W_k v_j \,\|\, W_k^e e'_{ji}])$$

The coefficients depend on the current edge embeddings, which carry the polarity and what the block has added to it. 113256 parameters, about 60% more.



the core block with a non-local block inserted

Attention or capacity: the two controls

GAT-Q-SAT adds the attention to the sum, so a gain may be due to the attention or simply to the larger network. Two more models, with the same configuration:

- Graph-Q-SAT wide: hidden layers of 104 units in place of 64, the capacity of GAT-Q-SAT with no attention;
- attention aggregation: the sum replaced by a mean weighted by attention (3 heads, averaged), at the capacity of Graph-Q-SAT.

	Graph-Q-SAT	Graph-Q-SAT wide	attention aggregation	GAT-Q-SAT
parameters	71144	113504	71627	113256
training seeds	3	2	2	3

- The port to the current stack
- The measure and the reference solver

Port and setup

The port, and a test of what it computes

- The code of 2020 is written for PyTorch 1.1 with pinned compiled extensions (`torch-scatter`, `torch-sparse`), which can no longer be installed: I ported it to the current PyTorch and PyTorch Geometric.
- The aggregations come from `torch_geometric.utils.scatter`; the environment runs with `gym` and `gymnasium`; the MiniSat wrapper is rebuilt against `numpy` 2; the checkpoints of 2021 load with the current `GATConv`.
- The checkpoints of 2021, evaluated on the new stack, give the same iterations as in 2021 on every instance in 63 of 86 evaluations, and medians within 0.03 in the others (on `flat30-60`: 1.833 and 2.000 on both stacks).
- The training path was checked by retraining the two coloring models from scratch (the seeds of 2026).

The measure and the reference solver

$$\mathrm{MRIR} = \mathrm{median}\ \frac{\#\text{iterations of MiniSat}}{\#\text{iterations of the model}}$$

over the instances of a family; an iteration is a decision, of the model or of VSIDS, which takes over after the cap (500 unless stated otherwise). Above 1 the model takes fewer decisions than MiniSat.

- The model drives MiniSat without restarts; the denominator is the stronger MiniSat on each family: with restarts on coloring (up to 26% fewer iterations), without on uniform random 3-SAT (2.2 to 2.9 times fewer at 250 variables).
- 100 test instances per family; graph coloring from 30 to 200 nodes, trained on 50.

- Graph coloring, by size
- Capacity or attention
- A change of the training distribution
- A change of domain
- Uniform random 3-SAT
- Iterations and running time

Results

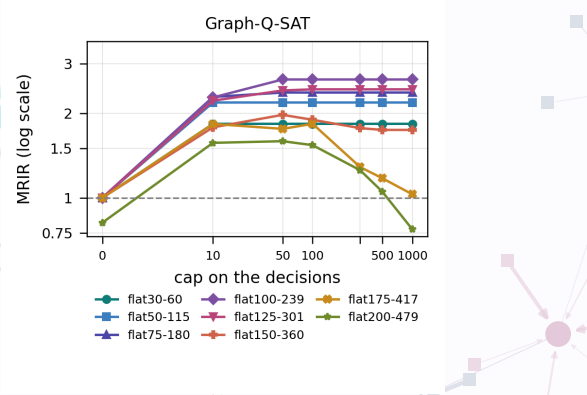
Graph coloring: the gap grows with the size

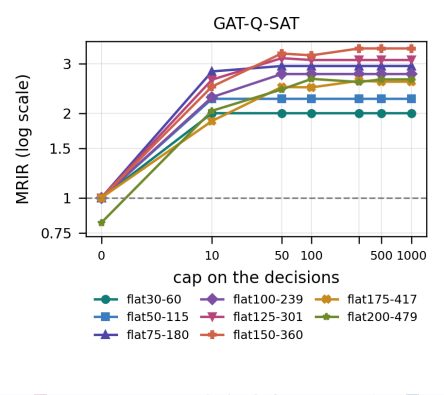
family	Graph-Q-SAT	GAT-Q-SAT	Δ
flat30-60	1.83	2.00	$+0.17$
flat50-115	2.18	2.25	$+0.07$
flat75-180	2.37	2.94	$+0.58$
flat100-239	2.64	2.76	$+0.12$
flat125-301	2.43	3.09	$+0.66$
flat150-360	1.74	3.40	$+1.66$
flat175-417	1.17	2.59	$+1.42$
flat200-479	1.05	2.64	$+1.59$

MRIR at cap 500, mean over the seeds

- GAT-Q-SAT takes fewer decisions on all eight families.
- Up to 100 nodes the two are close; from 125 nodes on the gap is large, mostly because Graph-Q-SAT degrades almost to the level of MiniSat at 200 nodes, four times the training size.

Graph coloring: the MRIR against the cap



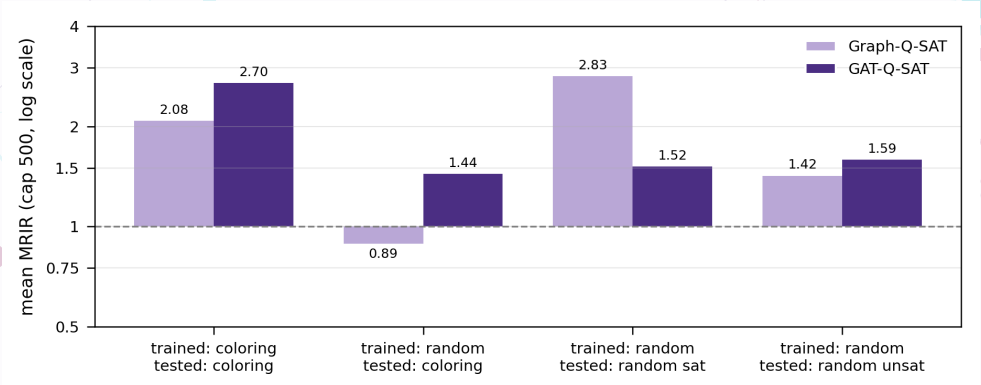


Runs of 2021, one line per family, each in its color (the same in every figure); the point at cap 0 is MiniSat without restarts. On the large families the decisions of Graph-Q-SAT after the first few tens make the search longer.

Capacity or attention

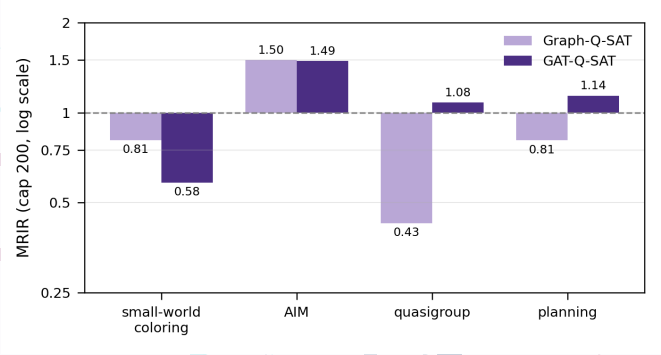
[The four variants on the eight families, from Table 2 of the note: whether Graph-Q-SAT wide closes the gap, whether attention aggregation reaches GAT-Q-SAT at a sixth of its added parameters, and the size of the differences against the variation across the seeds.]

Trained on random formulas, tested on graph coloring



Light lilac Graph-Q-SAT, deep violet GAT-Q-SAT. Trained on uniform random 3-SAT, Graph-Q-SAT is below MiniSat on five of the eight coloring families, GAT-Q-SAT above it on seven: the weighting of the neighbors helps also on a structure not seen in training.

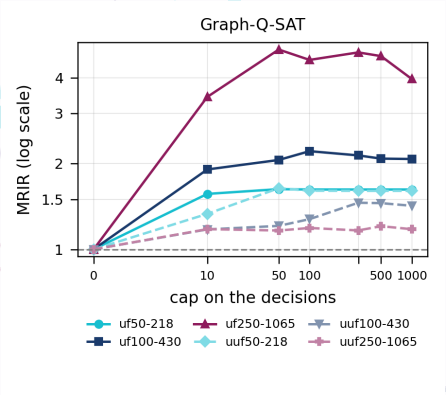
A change of domain: thirteen SATLIB families

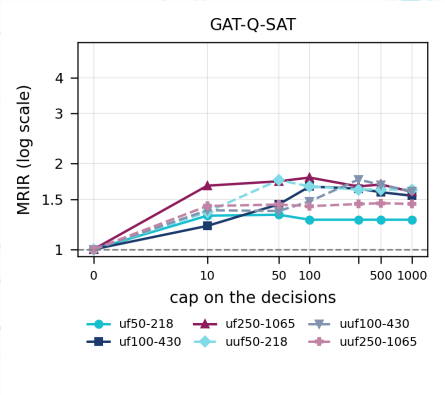


The coloring-trained models, with no retraining, on all the structured SATLIB families up to 2000 variables and 150000 clauses, and on small-world coloring at nine levels of rewiring; cap 200, MRIR against the stronger MiniSat.

[On how many families each variant is above MiniSat and above the others, and where the ranking flips.]

Uniform random 3-SAT: the ranking is reversed





On satisfiable formulas Graph-Q-SAT is better at all sizes (4.78 against 1.69 at 250 variables): at the phase transition the neighbors of a node are statistically alike. On unsatisfiable formulas the two are level or GAT-Q-SAT is ahead.

Iterations and running time

	cap 50		cap 500		cap 1000	
	sec	MRIR	sec	MRIR	sec	MRIR
Graph-Q-SAT	2.55	1.59	26.51	1.05	57.67	0.77
GAT-Q-SAT	3.52	2.43	12.20	2.64	12.19	2.64

`flat200-479`, median seconds per instance

- A decision of GAT-Q-SAT costs more, but at larger caps it is the faster of the two, since it terminates in fewer decisions.
- Both take seconds where MiniSat takes a fraction of a second: at this size neither is competitive in time, which is the motivation of the restricted heuristics (implemented, not measured) and of offline inference.

Conclusions and limits

- On graph coloring GAT-Q-SAT took fewer decisions on every family, with a gap that grows with the size; it was also the better of the two when trained on random formulas and tested on coloring.
- [The controls: attention or capacity.]
- On satisfiable uniform random 3-SAT Graph-Q-SAT was clearly superior, where there is no structure for the attention to weight.
- Limits: three seeds per model (two per control) do not resolve differences of a few tenths; neither model is competitive with MiniSat in time; the largest SATLIB instances (e.g., bounded model checking) are beyond the size I could evaluate.

Code, instances and logs: github.com/dmeoli/GQSAT.

Thank you for your *attention*!

github.com/dmeoli/GQSAT